

Emotion-aware Hybrid LLM Music Recommendation System

Goal

This project aims to build an emotion-aware music recommendation system that goes beyond traditional acoustic or co-listening signals. Instead of collaborative filtering, we use lyric's emotional semantics to better align recommendations with intended vibe.

Objective

We will build a dynamic hybrid recommendation system that combines playlist-based collaborative filtering (CF) with lyric-based emotion analysis based on LLM semantic extraction. The system first learns long-term playlist preference from song co-occurrence patterns, then dynamically adjusts recommendation weights based on recent emotional shifts within the playlist.

Problem

Current commercial systems (Spotify/Apple Music) primarily rely on Collaborative Filtering (user-item interaction) and low-level basic audio features (tempo, energy). These systems suffer from:

- **Semantic Blindness:** An inability to distinguish nuanced emotional narratives (e.g., "triumphant" vs. "aggressive" high-energy tracks) that are only evident in lyrics.
- **Cold-Start Sensitivity:** New or niche songs often lack the massive play history required for effective collaborative filtering, despite possessing the perfect "emotional vibe" for a user.

- **The Static Preference Fallacy:** Standard algorithms often assume user preferences remain static during a session, failing to detect mid-playlist mood shifts.

The Solution

Our solution implements a hybrid pipeline that combines collaborative filtering with LLM-based lyrical emotion analysis. By dynamically weighting long-term preferences against recent mood signals, the system can predict the "next song" with higher emotional accuracy and contextual relevance.

Novelty Justification

The project's novelty is grounded in three major, main-track conference publications from the past three years:

- **Main Research (LEMON) [1]:** This publication provides the core framework for hierarchical emotional extraction. We adopt its methodology for parsing lyrics into multi-layered emotional semantics and, more importantly, its approach to modeling short-term mood transitions. This allows our system to detect when a playlist's vibe shifts and adjust recommendations accordingly, addressing the "Static Preference" fallacy.
- **Supporting Research (KAR) [2]:** This work justifies the use of LLMs for Knowledge Augmentation. It demonstrates that recommendation systems can be improved by injecting open-world knowledge, such as the cultural or historical significance of a song, which is missing from raw user-item interaction matrices. We use this to bridge the gap between simple song IDs and the deeper contextual meaning of music.

- **Supporting Research (LLM-Rec) [3]:** This paper provides the methodological rigor for our implementation. It offers specific strategies for prompt engineering to extract high-fidelity semantic features from textual data. Furthermore, it establishes the evaluation protocol for assessing how well LLM-generated summaries and features correlate with actual user preferences, ensuring our LLM-as-a-Judge qualitative metrics are grounded in current research standards.

Dataset

- **Spotify Million Playlist Dataset (MPD):** Contains playlist titles, track lists, and artist/album/track metadata; used as the primary user-preference source where playlists serve as proxy user profiles.
- **Spotify Million Song Dataset:** Contains song titles, artist names, links, and lyrics; used to match MPD tracks with lyrical content, enabling the extraction of the semantic features necessary for emotion-based ranking.

Proposed Approach

- **Representation Learning:** First, we utilize the Spotify MPD to learn playlist-level song co-occurrence patterns, establishing a baseline for general user preferences.
- **Semantic Extraction:** Second, we utilize a low-parameter open-source LLM (e.g., Llama-3 or Gemini 1.5) to assign emotional embeddings to tracks. This allows for a superior interpretation of metaphor, indirect emotional expression, and multilingual lyrics.
- **Temporal Analysis:** Third, the system analyzes the first 80% of a playlist to detect emotional divergence points, identifying when a playlist shifts from one emotional style

(e.g., calm) to another (e.g., energetic). We estimate the average emotional switching point or switching frequency within the observed playlist.

- **Hybrid Prediction:** Finally, the model combines the long-term preference score with the short-term mood signal to generate a re-ranked list for the subsequent songs in the sequence.

Evaluation Metrics

The system will be evaluated using a leave-last-k-out strategy, using the first 80% of each playlist as input to predict the remaining 20% of left-out test songs. Performance will be measured via Recall@10 and NDCG@10 to determine if the emotion-aware signals provide a statistically significant lift over standard collaborative filtering.

Limitations

The primary limitation is the lack of real individual listening histories; treating playlists as proxy users may capture archival behavior rather than real-time states. Additionally, the MPD reflects behavior from 2010–2017, meaning the combined dataset may not fully represent modern music trends or current artist landscapes.

Ethics

To mitigate the risk of emotional spiraling, where a distressed user is repeatedly fed negative content, our system will incorporate diversity constraints to ensure the recommender does not trap users in a singular emotional category. Furthermore, we acknowledge that LLMs may carry cultural or linguistic biases, potentially misinterpreting irony, slang, or culturally specific lyrical metaphors.

Reference List

- [1] Shijia Wang, Tianpei Ouyang, Yunfan Zhou, Qiang Xiao, Yintao Ren, Yifei Pan, Fangjian Li, and Chuanjiang Luo. 2025. Enhanced Emotion-aware Music Recommendation via Large Language Models. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '25)*. Association for Computing Machinery, New York, NY, USA, 4986–4994. <https://doi.org/10.1145/3711896.3737212>
- [2] Yunjia Xi, Weiwen Liu, Jianghao Lin, Xiaoling Cai, Hong Zhu, Jieming Zhu, Bo Chen, Ruiming Tang, Weinan Zhang, and Yong Yu. 2024. Towards Open-World Recommendation with Knowledge Augmentation from Large Language Models. In *Proceedings of the 18th ACM Conference on Recommender Systems (RecSys '24)*. Association for Computing Machinery, New York, NY, USA, 12–22. <https://doi.org/10.1145/3640457.3688104>
- [3] Hanjia Lyu, Song Jiang, Hanqing Zeng, Yinglong Xia, Qifan Wang, Si Zhang, Ren Chen, Chris Leung, Jiajie Tang, and Jiebo Luo. 2024. LLM-Rec: Personalized Recommendation via Prompting Large Language Models. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 583–612, Mexico City, Mexico. Association for Computational Linguistics. <https://aclanthology.org/2024.findings-naacl.39/>
- [4] Himanshu Wagh. 2020. Spotify Million Playlist Dataset. Kaggle. <https://www.kaggle.com/datasets/himanshuwagh/spotify-million>
- [5] Shrirang. 2022. Spotify Million Song Dataset with Lyrics. Kaggle. <https://www.kaggle.com/datasets/notshrirang/spotify-million-song-dataset>